

Nonlinear Dynamic Inversion (NDI) Controller Design for a Wheeled Inverted Pendulum

1 Equations of Motion

Recall the state-space model for the wheeled inverted pendulum:

$$a \ddot{x} + b \cos \theta \ddot{\theta} - b \sin \theta \dot{\theta}^2 = F, \quad (1)$$

$$b \cos \theta \ddot{x} + c \ddot{\theta} - m_p g l \sin \theta = 0, \quad (2)$$

with the parameters defined as:

$$a = 2m_w + m_p + \frac{2I_w}{r^2}, \quad b = m_p l, \quad c = m_p l^2 + I_p.$$

Define the state vector as:

$$x = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix}.$$

Our objective is to choose a control input F such that the closed-loop system drives both the horizontal position x and the pendulum angle θ to zero.

2 Nonlinear Dynamic Inversion (NDI) Controller Design

The NDI approach aims to cancel the nonlinearities in the dynamics so that the overall system behaves like a linear system, where standard linear control techniques can then be applied.

2.1 Equations in Matrix Form

Express the dynamics in the form:

$$M(q) \ddot{q} + C(q, \dot{q}) = U,$$

where the generalized coordinate is

$$q = \begin{bmatrix} x \\ \theta \end{bmatrix}, \quad U = \begin{bmatrix} F \\ v \end{bmatrix}.$$

The inertia matrix is given by:

$$M(q) = \begin{bmatrix} a & b \cos \theta \\ b \cos \theta & c \end{bmatrix},$$

and the vector $C(q, \dot{q})$ contains the coupling nonlinear and gravitational terms.

2.2 Defining Auxiliary Inputs

Two auxiliary inputs are chosen:

- For the pendulum dynamics, set the auxiliary input v via a PD law to drive θ toward zero:

$$v = -K_{p\theta} \theta - K_{d\theta} \dot{\theta}.$$

- For the horizontal motion, define the desired horizontal acceleration using another PD law:

$$\ddot{x}_{des} = -K_{px} x - K_{dx} \dot{x}.$$

2.3 Dynamic Inversion and Back-Solving for F

The strategy is to invert the nonlinear dynamics using:

$$\begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} = M^{-1}(q) (U - C(q, \dot{q})),$$

and then “back-solve” for F such that the desired horizontal acceleration is achieved.

Enforce the following conditions:

- For the cart dynamics, we desire:

$$\ddot{x} = \ddot{x}_{des}.$$

- For the pendulum dynamics, we set:

$$\ddot{\theta} = v.$$

Multiplying by $M(q)$ gives:

$$M(q) \begin{bmatrix} \ddot{x}_{des} \\ v \end{bmatrix} = \begin{bmatrix} F \\ * \end{bmatrix} + C(q, \dot{q}).$$

Thus, by taking the first element of the above equation, the control input is obtained as:

$$F = \left[M(q) \begin{bmatrix} \ddot{x}_{des} \\ v \end{bmatrix} - C(q, \dot{q}) \right]_1.$$

This equation determines the required force F to ensure that the horizontal acceleration matches the desired $\ddot{x}_{des}$ while the pendulum is stabilized via v .

3 Tuning Guidelines

The performance of the NDI controller depends on the proper tuning of several parameters:

- **PD Gains for Pendulum ($K_{p\theta}$ and $K_{d\theta}$):** These gains control how aggressively the pendulum angle θ is driven toward zero. Increasing $K_{p\theta}$ reduces steady-state error, while increasing $K_{d\theta}$ improves damping but may increase control effort.
- **PD Gains for Horizontal Motion (K_{px} and K_{dx}):** These determine the desired horizontal acceleration. Higher gains result in faster response but might lead to overshoot if too aggressive.
- **Model Accuracy:** The inversion relies on accurate knowledge of $M(q)$ and $C(q, \dot{q})$. Any mismatch between the model and the true dynamics will affect performance. Robustification techniques (such as adaptive elements) may be incorporated if needed.

4 Simulation and Implementation

A typical simulation proceeds as follows:

1. Compute $\ddot{x}_{des}$ using the horizontal PD law.
2. Compute v using the pendulum PD law.
3. Form the desired acceleration vector:

$$\text{desired_acc} = \begin{bmatrix} \ddot{x}_{des} \\ v \end{bmatrix}.$$

4. Calculate the nonlinear terms $C(q, \dot{q})$ based on current q and $\dot{q}$.
5. Invert the dynamics:

$$\begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} = M^{-1}(q) \left(\begin{bmatrix} F \\ v \end{bmatrix} - C(q, \dot{q}) \right).$$

6. Solve for F by back-solving the first element:

$$F = \left[M(q) \text{desired_acc} - C(q, \dot{q}) \right]_1.$$

7. Integrate the system dynamics over time (using ODE solvers such as `ode45`) with initial conditions (e.g., a small initial deviation in θ).

5 Conclusion

The NDI controller cancels the inherent nonlinearities of the wheeled inverted pendulum by dynamically inverting the system's dynamics. By defining two auxiliary control inputs:

- $v = -K_{p\theta} \theta - K_{d\theta} \dot{\theta}$ (for pendulum stabilization),
- $\ddot{x}_{des} = -K_{px} x - K_{dx} \dot{x}$ (for horizontal motion),

and back-solving the dynamic equations, the control input F is computed to drive the system states x and θ to zero. Careful tuning of the PD gains and ensuring model accuracy are critical for achieving robust stabilization.